

YUNFAN KANG

Visiting Research Scientist, University of Illinois Urbana-Champaign

Email: yfkang@illinois.edu

Mobile: (+1) 515-996-0425

Address: Room 237E, Davenport Hall, UIUC

Google Scholar Page: <https://scholar.google.com/citations?user=T8vfAJIAAAAJ>

Linkedin: <https://www.linkedin.com/in/yunfan-kang/>

Work Experience

Sep 2025 - Present University of Illinois Urbana-Champaign

Visiting Research Scientist, CyberGIS Center for Advanced Digital and Spatial Studies

Nov 2023 - Sep 2025 University of Illinois Urbana-Champaign

Postdoc Research Associate, CyberGIS Center for Advanced Digital and Spatial Studies

Education and Training

Sep 2018 - Sep 2023 University of California, Riverside (UCR)

Ph.D. in Computer Science and Engineering (GPA 3.82/4)

Research Focus: Spatial Analysis, Big Spatial Data Management

Sep 2014 - Jun 2018 The University of Hong Kong

Bachelor of Engineering in Computer Science (Major GPA 3.837/4.3)

Minor in Mathematics (Minor GPA: 3.75/4.3)

Jul 2016 - Aug 2016 Peking University Summer School

Sep 2016 - Dec 2016 The University of California, San Diego

Exchange student (GPA: 3.825/4)

Research and Project Experience

Nov 2023 - Present PostDoc Research Associate and Visiting Research Scientist (UIUC)

- Supervised by [Prof. Shaowen Wang](#).
- Lead developer for the I-GUIDE platform User Environment, focusing on GeoAI and Generative AI, backend, search.
- Lead the research group Spatial, temporal, and network analysis
 - Supervised 2 undergraduate students and 1 graduate student.
 - Developed NetPointLib for large-scale spatial network point data fusion and analysis, presented at PEARC 2024
- Lead the research group of Geospatial Generative AI and Analysis
 - Supervised 1 undergraduate student, 2 Ph.D. students, and 1 postdoc.
 - One paper accepted by TKDE and one paper accepted by TSAS

Sep 2018 - Sep 2023 Graduate Research Assistant (UCR)

- Supervised by [Prof. Amr Magdy](#).
- Pyneapple: Scalable and expressive spatial analysis library.
Demo paper accepted for ICDE 2024.

- Coordinator of the library development and the demonstration development
- Statistical robust hotspot detection on spatial networks
Best paper candidate at SIGSPATIAL 23.
Invited extension submission to TSAS special issue.
- Enriched map-p regionalization algorithm
Presented at ICDE 22.
Incremental update method published in TKDE
- Unified Data Model for High-definition Map Data (ICDEW 20)
- Scalable Multi-resolution Spatial Visualization Database for Anthropogenic Litter Data (SIGSPATIAL 19)

Sep 2021 – Sep 2023 Teaching assistant (UCR)

- Spatial Computing Sharing academic and professional information.
- CS 10C Introduction to Data Structures and Algorithms
Taught the lecture about sorting algorithms in Nov 2022

Undergraduate Research Fellowship Program (HKU)

- Supervised by [Prof. Reynold C.K. Cheng](#).
- **Jun 2017 – Aug 2017 Summer research internship**
Implemented *C-Explorer*, which is a web-based platform facilitating the visualization and analysis of community search and community detection algorithms.
C-Explorer was demoed at the VLDB 2017 conference and was asked to be deployed online and made open source.
- **Sep 2017 – Mar 2018 Final year project**
Community search algorithms on the graph with edge labels.
Published in ICBDR 2018

Apr 2017 - Jun 2017 Student Research Assistant (HKU)

- Worked with Prof. Reynold C.K. Cheng and Big Data Transportation Team on the topic of *Enhancing Automatic Incident Detection with Social Media*.
- Did literature review and presentations on algorithms for recalculating navigation paths dynamically after an incident has been detected.

Tools and Skills

Programming languages: Java, Python, JavaScript, HTML, C++

Tools and techniques: QGIS, LaTeX, Overleaf, SciLab, GitHub, Anaconda, Jupyter Notebook, OmniGraffle, Node.js, Express.js, OpenSearch, LangChain, RAG

Packages: PyTorch, scikit-learn, GDAL, or GeoPandas

Leadership Experience and Other Projects

May 2024 ICDE 2024 Session Chair

Apr 2023 ICDE 2023 Volunteer Coordinator

Mar 2023 Student Assistantship - 2023 Esri Developer Summit

Sep 2019 - Jan 2020 The Science Fair Competition mentorship

- Mentored a student from Martin Luther King High in Riverside to work on a machine learning-based signature evaluation system. The project ranked third in the competition.

Sep 2016 - Nov 2016 *Let us do advanced mathematics project*

- Formed an entrepreneur team with students from Beihang University.
- Responsible for the web development and the app development.
- The project won several prizes, and a company was launched.

Sep 2015 - Jun 2018 HKU Mentorship program mentor

Sep 2015 HKU USEE new student orientation team leader

Sep 2015 - May 2016 Strait Talk Hong Kong – IT Secretary

- Responsible for maintaining the official website and technical problems at public events.
- Organized and hosted several public events and speeches.
- Cooperated with other secretaries to organize the annual STHK forum.

Aug 2015 Built the website for the HKU Mental Research Lab

External reviewer for SSTD, VLDB, ICDE, SIGMOD, SIGSPATIAL, BigData, Knowledge and Information Systems, and Scientific Reports, Social Network Analysis and Mining...

BigData 2025 PC member

KDD 2026 AI4Science Track Reviewer

SIGSPATIAL 2026 PC member

Publications

Under Review or Preprints

-
- Jimeng Shi, Runchu Tian, Wei Hu, Bowen Jin, Wonbin Kweon, Seongku Kang, Yunfan Kang, Dingqi Ye, Sizhe Zhou, Shaowen Wang and Jiawei Han. Multi-cube RAG for Multi-hop Question-Answering (Under review for VLDB 2026)
- Fangzheng Lyu, Yan Song, Xinlin Ma, Alexander Michels, **Yunfan Kang**, Shaowen Wang. Quantifying Urban Retrofitting: Where, When and What (Under review for Cities)
- Elizabeth G Campolongo, Yuan-Tang Chou, Ekaterina Govorkova, Wahid Bhimji... **Yunfan Kang**. *Building Machine Learning Challenges for Anomaly Detection in Science*. *arXiv preprint arXiv:2503.02112*, 2025.

Journal Papers

- Fangzheng Lyu, **Yunfan Kang**, Shaowen Wang. Synthesizing high-performance computing and machine learning for geospatial knowledge discovery - a comprehensive review and research agenda, *Annals of GIS*, 1–22.
- **Yunfan Kang**, Yiyang Bian, Qinma Kang, and Amr Magdy, IMS: Incremental Max-P Regionalization with Statistical Constraints, *IEEE Transactions on Knowledge and Data Engineering*, 2025.
- **Yunfan Kang**, Yongyi Liu, Pratham Juvekar, Ahmed R. Mahmood, Shaowen Wang, Amr Magdy, Progressive and Scalable Hotspot Detection through Local K-Function in Spatial Networks. *ACM Transactions on Spatial Algorithms and Systems*, 2025.
- Yiran Chen, Qinma Kang, Wenqiang Duan, Yunfan Shan, Ran Xiao, **Yunfan Kang**. *An Iterated Local Search Algorithm for Community Detection in Signed Networks*. *International Journal of Modern Physics C*, 33(08), 2250105, 2022.
- Wenqiang Duan, Qinma Kang, **Yunfan Kang**, Jianwen Chen, Qingfeng Qin. *A Simple and Effective Iterated Greedy Algorithm for Structural Balance in Signed Networks*. *International Journal of Modern Physics B*, 36(21), 2250129, 2022.

- Hannah Hapich, Win Cowger, Andrew Gray, Neil Tangri, Tony Hale, Amr Magdy, Antoinette Vermilye, **Yunfan Kang**, et al. *Trash Taxonomy Tool: Harmonizing Classification Systems Used to Describe Trash in Environments*. *Microplastics and Nanoplastics*, 2(1), 15, 2022.
- Abdulaziz Almaslukh, **Yunfan Kang**, Amr Magdy. "Geo-Social Personalized Keyword Search Over Streaming Data." *ACM Transactions on spatial algorithms and systems* (2021).
- Chao Liu, Qinma Kang, Hanzhang Kong, Wenquan Li, **Yunfan Kang**. An interdependent network coupling strategy based on overlapping link structure against targeted attack[J]. *International Journal of Modern Physics C*, 2021: 2150101.
- Yunfan Shan, Qinma Kang, Ran Xiao, Yiran Chen, **Yunfan Kang**. *An Iterated Carousel Greedy Algorithm for Finding Minimum Positive Influence Dominating Sets in Social Networks*. *IEEE Transactions on Computational Social Systems*, 9(3), 830–838, 2021.
- Wenquan Li, Qinma Kang, Hanzhang Kong, Chao Liu, **Yunfan Kang**. A novel iterated greedy algorithm for detecting communities in complex network[J]. *Social Network Analysis and Mining*, 2020, 10: 1-17.
- Chao Liu, Qinma Kang, Hanzhang Kong, Wenquan Li, **Yunfan Kang**. An iterated local search algorithm for community detection in complex networks. *International Journal of Modern Physics B* 34 (04), 2050013, 2020
- Hanzhang Kong, Qinma Kang, Wenquan Li, Chao Liu, **Yunfan Kang**, Hong He, 2018. A hybrid iterated carousel greedy algorithm for community detection in complex networks, *Physica A: Statistical Mechanics and its Applications*, 2019 Volume 536
- Amr Magdy, **Yunfan Kang***, Laila Abdelhafeez*, Eric Ong, and Mohamed F. Mokbel. Microblogs Data Management: A Survey. To appear in the *International Journal on Very Large Data Bases*, *VLDB Journal*, 2019. (* equal contributions)
- Hanzhang Kong, Qinma Kang, Chao Liu, Wenquan Li, Hong He, **Yunfan Kang**. (2018). An improved label propagation algorithm based on node intimacy for community detection in networks. *International Journal of Modern Physics B*, 32(25), 1850279.

Conference Papers

- Shaowen Wang, Furqan Baig, **Yunfan Kang**, Erick Li, Alexander Michels, Nattapon Jaroenchai, Anand Padmanabhan, Arunesh Kumar, Rajesh Kalyanam, Noah S. Oller Smith, X. Carol Song, Lan Zhao, Steven Manson, Mohan Ramamurthy, Sina Taghavikish and David Tarboton, I-GUIDE Platform for Geospatial Data-intensive Convergence Research and Education, *PEARC 2025*
- Fangzheng Lyu, Furqan Baig, **Yunfan Kang**, Zhiyuan Li, Anand Padmanabhan and Shaowen Wang. A Convergence Framework for Integrating CyberGIS Education and Research on I-GUIDE Platform. *I-GUIDE Forum 2024*.
- **Yunfan Kang**, Fangzheng Lyu, Shaowen Wang, NetPointLib: Library for Large-Scale Spatial Network Point Data Fusion and Analysis, *PEARC 2024*
- Yongyi Liu*, **Yunfan Kang***, Ahmed Mahmood, Amr Magdy. Scalable Evaluation of Local K-Function for Radius-Accurate Hotspot Detection in Spatial Networks. *SIGSPATIAL 2023* (* equal contributions)
- **Yunfan Kang**, and Amr Magdy. "EMP: Max-P Regionalization with Enriched Constraints." *2022 IEEE 38th International Conference on Data Engineering (ICDE)*. IEEE, 2022.
- Qinma Kang, **Yunfan Kang**, Hanzhang Kong. 2018. Edge-Attributed Community Search for Large Graphs. In *Proceedings of the 2nd International Conference on Big Data Research*, (pp. 114-118).
- Qinma Kang, Huizhuo Zhou, **Yunfan Kang**. 2018. An Asynchronous Advantage Actor-Critic Reinforcement Learning Method for Stock Selection and Portfolio Management. In *Proceedings of the 2nd International Conference on Big Data Research (ICBDR 2018)*. Association for Computing Machinery, New York, NY, USA, 141–145.

System Demonstrations

- Yongyi Liu, Nicolas Lee, **Yunfan Kang**, Mohammad Reza Shahneh, Ahmed Mahmood, Vishal Rohith Chinnam, Aparna Vivek Sarawadekar, Samet Oymak, Ibrahim Sabek and Amr Magdy. Pyneapple-L: Scalable Expressive Learning-based Spatial Analysis. In *Proceedings of the 32nd ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (2024)*.
- **Yunfan Kang**, Yongyi Liu, Hussah Alrashid, Akash Bilgi, Siddhant Purohit, Ahmed Mahmood, Sergio Rey, Amr Magdy. Pyneapple-R: Scalable and Expressive Spatial Regionalization. *2024 IEEE 40th International Conference on Data Engineering (ICDE)*. IEEE, 2024.
- **Yunfan Kang**, Ziang Zhao, Amr Magdy, Win Cowger, Andrew Gray. Scalable Multi-resolution Spatial Visualization for Anthropogenic Litter Data. In *Proceedings of the 27th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems* (pp. 560-563).

Workshop Papers

- **Yunfan Kang**, Amr Magdy. HiDaM: A Unified Data Model for High-definition (HD) Map Data[C]//*2020 IEEE 36th International Conference on Data Engineering Workshops (ICDEW)*. IEEE, 2020: 26-32.
- Ziang Zhao, **Yunfan Kang**, Amr Magdy, Win Cowger, Andrew Gray. A Data-Driven Approach for Tracking Human Litter in Modern Cities. In *2019 IEEE 35th International Conference on Data Engineering Workshops (ICDEW)* (pp. 69-73). IEEE.

Poster and Presentations

- **Yunfan Kang**, Ziang Zhao, Amr Magdy, Win Cowger, Andrew Gray. Scalable Multi-resolution Spatial Visualization for Anthropogenic Litter Data. *32nd ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*.
- **Yunfan Kang**, Yongyi Liu, Hussah Alrashid, Akash Bilgi, Siddhant Purohit, Ahmed Mahmood, Sergio Rey, Amr Magdy. Pyneapple-R: Scalable and Expressive Spatial Regionalization. *40th IEEE International Conference on Data Engineering*.
- **Yunfan Kang**, Zhiyuan Li, Furqan Baig, Anand Padmanabhan, Nattapon Jaroenchai and Shaowen Wang. Democratizing Convergence Science through Open Search and API Capabilities of I-GUIDE Platform. *I-GUIDE Forum 2024*.
- Furqan Baig, **Yunfan Kang**, Zhiyuan Li, Anand Padmanabhan, Nattapon Jaroenchai and Shaowen Wang. Graph-based Knowledgebase of I-GUIDE Platform for Empowering Data-intensive Convergence Science. *I-GUIDE Forum 2024*.
- Erick Li, Nattapon Jaroenchai, Yunfan Kang, Furqan Baig, Anand Padmanabhan and Shaowen Wang. User Interface of I-GUIDE Platform for Convergent Knowledge Sharing and Discovery. *I-GUIDE Forum 2024*.
- **Yunfan Kang**, Erick Li, Furqan Baig, Anand Padmanabhan and Shaowen Wang. AI-Driven Spatial, Semantic, and Full-Text Search: An Integrated Framework for the I-GUIDE Platform. *I-GUIDE Forum 2025*.
- Erick Li, Furqan Baig, **Yunfan Kang**, Rohan Colaco, Pierre Hanlet, Nattapon Jaroenchai, Anand Padmanabhan and Shaowen Wang. I-GUIDE Platform User Environment for Enabling Geospatial Data-intensive Knowledge Discovery and Sharing. *I-GUIDE Forum 2025*.
- Furqan Baig, Arunesh Kumar, **Yunfan Kang**, Erick Li, Anand Padmanabhan and Shaowen Wang. Bridging Data and Discovery: A Scalable Knowledge Graph Platform with Object Storage and Administrative Oversight. *I-GUIDE Forum 2025*.
- Erick Li, Arunesh Kumar, Furqan Baig, **Yunfan Kang**, Nattapon Jaroenchai, Anand Padmanabhan and Shaowen Wang. I-GUIDE Platform for Geospatial Data-Intensive Knowledge

Discovery through Scalable Knowledge Graph and Object Storage. Gateways 2025 Annual Conference.

Tutorials

- Enhancing Geospatial-data Intensive Knowledge Discovery with OpenSearch and LLM Search I-GUIDE Forum 2025

Honors and Awards

- 2024 NSF HDR NextGen Leaders Fellowship Award.
- SIGSPATIAL 2019 Student Travel Awards.
- 2018 Deans Distinguished Fellowship. UC, Riverside.
- 2018 First Class Honours Graduation, HKU.
- 2018 Third runner-up of the best final year project in the CS department of HKU.
- 2018 Dean's Honour List, HKU.
- 2016 Third prize in The Activity "Undergraduates Making Innovations in Yinchuan".
- 2016 Philip K H Wong Foundation Scholarships for Student Enrichment.
- 2015 First prize in the National Mobile Internet Innovation Contest.

Professional Affiliations

Alumni of Center for GeoSpatial Sciences, University of California, Riverside

Member of CyberGIS Center for Advanced Digital and Spatial Studies, University of Illinois Urbana-Champaign

Project Contribution

CAREER: Scalable Spatial Data Science on User-generated Data. Award number 2237348

HDR Institute: Geospatial Understanding through an Integrative Discovery Environment. Award number 2118329